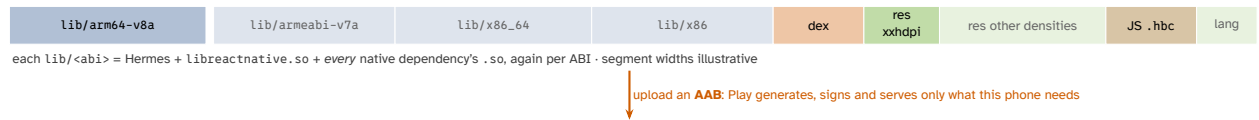


In one line: Users download a **per-device slice**: from an **AAB**, Play builds **split APKs** (base + one **ABI** + one **density** + the device's **languages**); the App Store thins iOS uploads. In RN the heavy part is **native libraries** × **ABIs** (Hermes, libreactnative.so, every native dependency), then assets. Measure *download* size.

universal APK — every device gets all of it



what a Pixel (arm64, xxhdpi, English) downloads



The levers — only wins a primary source states are quantified

lever	what it removes / how	stated win
AAB (required for new apps, Aug 2021)	other ABIs, densities, languages — from the <i>download</i>	per device
R8 minify + shrinkResources	unreachable bytecode + unused resources; short names	—
Hermes (default engine)	Android: no JSC library to ship (RN docs: “smaller app size”)	—
RN 0.76 merged libreactnative.so	code duplicated across many .so files	~3.8 MB (~20%)
RN 0.84 on iOS	legacy-architecture code no longer compiled in	“reduces app size”
remove unused native dependencies	their .so/pods — autolinking links <i>every</i> native package	—
WebP images, vector drawables	image bytes; one XML instead of a PNG per density	—
resourceConfigurations	translations bundled by libraries that you do not ship	—

How it works

- **ABIs:** the template builds armeabi-v7a, arm64-v8a, x86, x86_64 (reactNativeArchitectures). With an AAB that costs *upload* size only. Outside Play (per-CPU APKs, splits { abi }) each APK needs its own **versionCode** — e.g. versionCode * 1000 + abiCode, arm64 higher so 64-bit devices prefer it.
- **R8** (enableProguardInReleaseBuilds in the RN template, off by default): tree-shakes, inlines, merges classes, obfuscates; full mode is default since AGP 8.0. shrinkResources only works with minify on.
- **Native libs packaging:** uncompressed, page-aligned .so (useLegacyPackaging = false) are mmap'd in place — no extraction, smaller install/updates; compressed libs are extracted at install (bigger on disk).
- **JS:** mostly dependencies — prune with a bundle visualizer.
- **iOS:** slicing + asset catalogs + ODR (see app-size-and-energy). JSC is a system framework there, so Hermes *adds* its engine to an iOS app. RN's require()d images are copied as loose files, not an asset catalog, so they are not sliced (*unverified*).

Example — Android release config

```
// android/app/build.gradle
def enableProguardInReleaseBuilds = true
android {
  defaultConfig { resourceConfigurations += ["en", "de"] }
  buildTypes { release {
    minifyEnabled enableProguardInReleaseBuilds
    shrinkResources true // needs minify
    proguardFiles getDefaultProguardFile(
      "proguard-android-optimize.txt"), "proguard-rules.pro"
    } }
}
// react-native.config.js: stop autolinking a lib on Android
// dependencies: { 'some-lib': { platforms: { android: null } } }
```

Measure — the number users see

- bundletool build-apks --bundle=app.aab --output=app.apks, then bundletool get-size total --apks=app.apks (per-config download sizes; --device-spec for one phone).
- Android Studio **APK Analyzer** (Build → Analyze APK): per-folder raw vs download size, dex references, .so alignment.
- **Play Console** app bundle explorer: size per device configuration.
- **iOS:** *App Thinning Size Report*. CI: fail past a size budget.

Traps

- ! Dropping x86/x86_64 “to shrink” an AAB — Play already serves one ABI; you only broke emulators and x86 devices.
- ! Quoting the universal APK or the upload size — users get the split.
- ! R8 full mode removes classes reached only by reflection/JNI → release-only ClassNotFoundException; add keep rules, test release.
- ! Stripping native symbols with nowhere to keep them — upload them (debugSymbolLevel) or crashes become unreadable.

Remember

Ship an AAB, minify + shrink, prune native deps, right-size assets — measure the download, not the upload.